

AI / ML / GenAI

10 Production-Grade Portfolio Case Studies

Business problem → solution → architecture → technology → measurable outcome

AI/ML Engineering
Production ML systems, forecasting, fraud, churn & monitoring

GenAI & Agentic AI
RAG, document intelligence, multi-agent workflows & copilots

MLOps
MLflow, CI/CD, serving, observability, governance & retraining

Portfolio context

These case studies demonstrate practical AI/ML, GenAI, Agentic AI and MLOps solution design through business problems, architecture, engineering decisions and measurable evaluation. They are structured to demonstrate production-oriented thinking and technical depth.

Important: The metrics in these case studies are illustrative portfolio benchmarks, not claims of results delivered for a specific client. Replace them with verified project metrics where available.

01 Predictive Demand Forecasting

From reactive planning to production-grade ML forecasting • Retail & Supply Chain

BUSINESS PROBLEM	SOLUTION
Planners were relying on spreadsheets and lagging indicators to forecast SKU-level demand. Promotions, intermittent demand and plant-level seasonality created frequent stock-outs and excess inventory.	Built a multi-stage forecasting platform that ingests sales and promotion signals, creates time-aware features in PySpark, trains champion/challenger models, and publishes forecasts through a governed MLflow registry. A business-facing dashboard exposes forecast, confidence bands and exception SKUs.

ARCHITECTURE

Sources	PySpark feature pipeline	Feature validation	Forecast models	MLflow Registry	Batch inference	Forecast API / BI
→ data flow / control flow →						
TECHNOLOGY	ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE		
Azure Databricks • PySpark • MLflow • Azure Data Lake • Delta Lake • Airflow • Power BI • Python	Illustrative portfolio benchmark: 18–25% lower WAPE on priority SKUs; 12% fewer stock-out exceptions; retraining cycle reduced from days to hours.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

Suggested screenshot: Create one UI frame showing the key business output (forecast chart, monitoring dashboard, cited RAG answer, incident timeline, reviewer queue, pipeline status, churn segment, fraud score, DQ scorecard or executive KPI answer).

02 Real-Time ML Monitoring & Drift

Detect model degradation before the business feels it • Financial Services

BUSINESS PROBLEM	SOLUTION
A production scoring service could remain technically healthy while its predictions became less reliable as customer behavior shifted. Existing monitoring focused on infrastructure uptime rather than model quality.	Designed an MLOps observability layer tracking data drift, prediction drift, feature nulls, latency, throughput and delayed ground-truth performance. Threshold breaches trigger alerts and an automated retraining workflow with approval gates.

ARCHITECTURE

Production Data	Validation	Feature Drift	Model / Prediction Drift	Metrics & Logs	Alerting	Retrain / Approve
→ data flow / control flow →						
TECHNOLOGY	ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE		
MLflow • Python • FastAPI • Docker • Kubernetes • Airflow • Azure Monitor • Evidently-style metrics • GitHub Actions	Illustrative portfolio benchmark: 60% faster detection of drift events; 35% reduction in manual model-health checks; automated retraining workflow shortened recovery from days to <1 business day.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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03 Enterprise RAG Knowledge Copilot

Turn scattered enterprise knowledge into grounded answers • Energy / Operations

BUSINESS PROBLEM	SOLUTION
Engineers and operations teams spent significant time searching manuals, procedures, incident reports and policy documents. Generic chatbots produced answers without reliable source grounding.	Built a retrieval-augmented generation workflow with document ingestion, chunking, metadata filtering, hybrid/vector retrieval, reranking and citation-aware response generation. Access control is enforced before retrieval.

ARCHITECTURE

Documents	Parse / Chunk	Embeddings	Vector + Keyword Search	Rerank	LLM	Cited Answer
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→ data flow / control flow →

TECHNOLOGY	ILLUSTRATIVE RESULTS	BEHANCE STORY ANGLE
Azure OpenAI / LLM • Azure AI Search / Vector Search • Python • LangChain • FastAPI • Azure Blob • Entra ID	Illustrative portfolio benchmark: 45% reduction in knowledge-search time; >90% of evaluated answers grounded in retrieved evidence; response trace available for audit.	Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.

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04 Multi-Agent Incident Response

Coordinate specialized AI agents instead of one oversized prompt • Cloud Operations

BUSINESS PROBLEM	SOLUTION
On-call engineers had to correlate alerts, logs, deployment changes and runbooks manually. A single general-purpose assistant struggled with tool selection and multi-step investigation.	Created an orchestrated agent system: an incident triage agent classifies the event, a telemetry agent gathers evidence, a change-analysis agent checks recent deployments, and a runbook agent proposes remediation. A supervisor controls tool access and escalation.

ARCHITECTURE

Alert	Supervisor	Triage Agent	Telemetry Agent	Change Agent	Runbook Agent	Human Approval
→ data flow / control flow →						
TECHNOLOGY		ILLUSTRATIVE RESULTS		BEHANCE STORY ANGLE		
Python • LLMs • LangGraph / agent orchestration • FastAPI • Kubernetes • OpenTelemetry • Azure Monitor		Illustrative portfolio benchmark: investigation workflow reduced from ~30 minutes to <10 minutes in test scenarios; 40% fewer manual diagnostic steps; human approval retained for remediation.		Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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05 GenAI Contract & Document Intelligence

Extract decisions, obligations and risk signals from unstructured documents • Legal / Procurement

BUSINESS PROBLEM	SOLUTION
Contract review required manual reading of long documents to identify renewal dates, payment terms, obligations, deviations and risk clauses.	Built a document intelligence pipeline that extracts structured fields, identifies clauses, compares them with policy rules and produces a review-ready summary. Low-confidence items are routed to humans.

ARCHITECTURE

PDF / Scans	OCR	Layout Parse	Clause Extraction	Policy Checks	Risk Scoring	Review UI
→ data flow / control flow →						
TECHNOLOGY	ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE		
Python • OCR / Document Intelligence • LLM • Pydantic • FastAPI • Azure Blob • Vector Search • React	Illustrative portfolio benchmark: 70% reduction in first-pass review time; >95% extraction accuracy on defined fields in test set; human review focused on exceptions.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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06 End-to-End MLOps Platform

Standardize the path from notebook experiment to governed production model • Enterprise Analytics

BUSINESS PROBLEM	SOLUTION
Data scientists were deploying models inconsistently, making reproducibility, approvals, versioning and rollback difficult.	Implemented a reusable ML lifecycle: Git branching, CI tests, data validation, experiment tracking, model registry, approval gates, containerized serving and scheduled retraining. Templates reduce project setup time.

ARCHITECTURE

Git	CI Tests	Databricks Jobs	MLflow Tracking	Registry	Container Build	Deployment
→ data flow / control flow →						
TECHNOLOGY	ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE		
Azure Databricks • MLflow • GitHub Actions • Docker • Kubernetes • Python • PySpark • Airflow	Illustrative portfolio benchmark: 50% faster project onboarding; reproducible model releases; rollback reduced to a controlled registry operation; deployment steps standardized across teams.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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07 Customer Churn & Next-Best-Action

Move from churn prediction to actionable retention • Telecom / Subscription

BUSINESS PROBLEM	SOLUTION
The business had a churn model but limited adoption because scores did not explain why customers were at risk or what action should be taken.	Combined propensity modeling with explainability and a next-best-action layer. Segments are scored daily, drivers are surfaced, and recommended interventions are sent to CRM workflows.

ARCHITECTURE

Customer Data	Feature Store	Churn Model	Explainability	NBA Rules	CRM	Outcome Feedback
→ data flow / control flow →						
TECHNOLOGY	ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE		
Python • Scikit-learn / XGBoost • SHAP • PySpark • MLflow • SQL • Power BI • CRM API	Illustrative portfolio benchmark: 15% lift in targeted retention response in offline simulation; 30% reduction in manual campaign selection; transparent reason codes for every recommendation.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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08 Fraud Detection & Real-Time Scoring

Score suspicious transactions in milliseconds • Payments

BUSINESS PROBLEM	SOLUTION
Batch fraud analysis created delays, while simple rules generated too many false positives. The business needed low-latency scoring with explainable signals.	Designed a streaming feature pipeline and real-time scoring API with layered rules + ML. The service returns risk score, top contributing signals and a decision recommendation.

ARCHITECTURE

Transaction	Stream	Feature Cache	Rules + ML	Risk Score	Decision	Case Management
→ data flow / control flow →						
TECHNOLOGY		ILLUSTRATIVE RESULTS		BEHANCE STORY ANGLE		
Python • FastAPI • Kafka/Event Hubs • PySpark Structured Streaming • XGBoost • Redis • Docker • Kubernetes		Illustrative portfolio benchmark: sub-100ms target scoring path in performance tests; 20% reduction in false-positive review volume; full decision trace retained.		Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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09 AI-Powered Data Quality & Reconciliation

Find broken data pipelines before reports break • Data & Finance

BUSINESS PROBLEM	SOLUTION
Daily reconciliations depended on manual SQL checks across source, warehouse and reporting layers. Failures were often discovered after downstream consumers complained.	Built automated data-quality checks for schema, freshness, volume, uniqueness, referential integrity and aggregate reconciliation. An AI assistant summarizes failures and links them to likely upstream causes.

ARCHITECTURE

Source	DQ Rules	Reconciliation	Anomaly Detection	LLM Summary	Alert	Scorecard
→ data flow / control flow →						
TECHNOLOGY		ILLUSTRATIVE RESULTS		BEHANCE STORY ANGLE		
PySpark • SQL • Databricks • Great Expectations-style checks • LLM • Airflow • Power BI • Azure		Illustrative portfolio benchmark: 80% of recurring checks automated; incident triage time cut by ~50%; data-quality scorecards available before business reporting windows.		Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.		

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10 Agentic Analytics & Executive Copilot

From natural-language questions to governed analytics • Enterprise Management

BUSINESS PROBLEM	SOLUTION
Executives and business users depended on analysts for recurring KPI questions. Direct LLM-to-database access created security and correctness concerns.	Created a governed analytics agent that interprets questions, maps them to approved semantic metrics, generates and validates SQL, executes read-only queries, checks anomalies, and returns a concise narrative with tables and drill-downs.

ARCHITECTURE

User Question	Intent	Metric Catalog	SQL Planner	SQL Validator	Query Engine	Narrative
→ data flow / control flow →						
TECHNOLOGY		ILLUSTRATIVE RESULTS			BEHANCE STORY ANGLE	
Python • LLM • Semantic Layer • SQL • Databricks / Synapse • FastAPI • React • Entra ID • Audit Logging		Illustrative portfolio benchmark: 60% reduction in repetitive KPI requests; >95% SQL validation pass rate on test suite; every answer logged with metric definition and query lineage.			Show the before/after workflow, architecture, one polished UI screen, and a short technical decision log. Focus on business value first, then engineering depth.	

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